

OLIVER HIXON-FISHER

Publication List | Human Motion Synthesis, Digital Humans & Embodied AI

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PUBLICATION LIST

Status current as of August 2026. Published/accepted outputs are separated from manuscripts under review and preprints to make publication status explicit.

Peer-reviewed publications and accepted work

Hixon-Fisher, O., Francik, J., & Makris, D. *High-Fiving with the Machine: Synthesising Reactive Motion from Human Input with Interaction Contact Labels*. 18th Annual ACM SIGGRAPH Conference on Motion, Interaction and Games (MIG 2025), ACM, long paper, Article 13, pp. 13:1-13:11, 2025, DOI - 10.1145/3769047.3769063

Hixon-Fisher, O., Francik, J., & Makris, D. *Motion Synthesis via Adaptive Guidance Experts*. In Computer Vision, Imaging and Computer Graphics Theory and Applications, Communications in Computer and Information Science, Springer Nature, 2025. Accepted / in press.

Hixon-Fisher, O., Francik, J., & Makris, D. *Pose-Centric Motion Synthesis Through Adaptive Instance Normalization*. Proceedings of the 20th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP 2025), Vol. 2: VISAPP, SCITEPRESS, pp. 39-47, 2025, DOI - 10.5220/0013102100003912

Manuscripts under review and preprints

Hixon-Fisher, O., Francik, J., & Makris, D. *Shaking Hands with the Machine - Reactive Motion Synthesis*. SSRN preprint, posted 4 June 2026. Extended journal version currently under review at Sensors (MDPI).

Hixon-Fisher, O. et al. *Sparse2Motion: Adaptive Full-Body Pose Reconstruction from VR Inputs*. Under review at Intelligent Systems with Applications (Elsevier), 2026.

Earlier research output

Hixon-Fisher, O. *A Combined GAN Approach Towards Procedural Generation in 3D Video Game Level Design*. Poster paper based on master's dissertation, 2nd International Conference on Interactive Media, Smart Systems and Emerging Technologies, 2022.

RESEARCH THEMES

Human motion synthesis; reactive interaction; digital humans and avatars; sparse-sensor full-body reconstruction; VR/XR sensing; diffusion models; JEPA/self-supervised learning; efficient sequence models; test-time adaptation and domain shift.